

Complements of Operational Research

1.2 Non-probabilistic decision criteria

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Deciding without probabilities

From 1.1: the **decision matrix**.

	θ_1	θ_2	$\cdots$	θ_m
a_1	c_{11}	c_{12}	$\cdots$	c_{1m}
a_2	c_{21}	c_{22}	$\cdots$	c_{2m}
$\vdots$	$\vdots$	$\vdots$		$\vdots$
a_n	c_{n1}	c_{n2}	$\cdots$	c_{nm}

a_i : actions (the decision maker controls them); θ_j :
states of nature (does not control them); c_{ij} :
consequence (here, a **gain**: the larger, the better).

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Question: if we know **nothing** about the probability of each θ_j , how do we choose an action?

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Each criterion summarises each row in a single number and chooses the best row.

Different criteria express different attitudes towards uncertainty.

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Non-probabilistic criteria

Each criterion summarises each row in a single number and chooses the best row.

Different criteria express different attitudes towards uncertainty.

Before applying any criterion: if $c_{ij} \leq c_{kj}$ for all j , action a_i is **dominated** by a_k and can be eliminated — no reasonable criterion chooses it.

Today's example

A company is going to install a new production line and has to choose its **capacity** before knowing the **demand**. Estimated annual profit (thousands of euros):

Capacity (actions)	Demand (states of nature)		
	low θ_1	medium θ_2	high θ_3
a_1 : small	40	45	50
a_2 : medium	20	60	80
a_3 : large	-40	50	120

Small line: almost certain profit, but it does not take advantage of a high demand. Large line: the largest possible profit, but a loss if demand is low. No action is dominated.

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We will apply five criteria — and see that they do not agree.

I. Maximax and II. Maximin

I. Maximax (optimistic criterion)

Assumes that the state **most favourable** to each action always occurs:

$$\max_{a_i} \left\{ \max_{\theta_j} c_{ij} \right\}$$

II. Maximin (pessimistic criterion)

Assumes that the state **least favourable** to each action always occurs:

$$\max_{a_i} \left\{ \min_{\theta_j} c_{ij} \right\}$$

I. Maximax and II. Maximin

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	θ_1	θ_2	θ_3	max.
a_1	40	45	50	50
a_2	20	60	80	80
a_3	-40	50	120	120

$$\max\{50, 80, 120\} = 120 \Rightarrow a_3 \text{ (large)}$$

II. Maximin (pessimistic criterion)

Assumes that the state **least favourable** to each action always occurs:

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II. Maximin (pessimistic criterion)

Assumes that the state **least favourable** to each action always occurs:

$$\max_{a_i} \left\{ \min_{\theta_j} c_{ij} \right\}$$

	θ_1	θ_2	θ_3	min.
a_1	40	45	50	40
a_2	20	60	80	20
a_3	-40	50	120	-40

$$\max\{40, 20, -40\} = 40 \Rightarrow a_1 \text{ (small)}$$

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a_1	40	45	50	40
a_2	20	60	80	20
a_3	-40	50	120	-40

$$\max\{40, 20, -40\} = 40 \Rightarrow a_1 \text{ (small)}$$

The optimist looks only at the **best** column of each row; the pessimist, only at the **worst**. Both ignore the rest of the row.

III. Laplace criterion

III. Laplace (insufficient reason)

With no reason to prefer one state to another, all states are given the same probability $1/m$:

$$\max_{a_i} \left\{ \frac{1}{m} \sum_{j=1}^m c_{ij} \right\}$$

Unlike criteria I and II, it uses the whole row.

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Unlike criteria I and II, it uses **the whole row**.

	θ_1	θ_2	θ_3	$\frac{1}{3} \sum_j c_{ij}$
a_1	40	45	50	$135/3 = 45.0$
a_2	20	60	80	$160/3 \approx 53.3$
a_3	-40	50	120	$130/3 \approx 43.3$

$$\max\{45.0, 53.3, 43.3\} = 53.3 \Rightarrow a_2 \text{ (medium)}$$

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Caution: the result depends on how the states are defined. If we split “high demand” into “high” and “very high”, the average changes — and with it, possibly, the decision.

IV. Savage criterion (minimax regret)

IV. Savage

1. In each column, compute the maximum $\max_k c_{kj}$.
2. **Regret** (or opportunity loss):
 $r_{ij} = \max_k c_{kj} - c_{ij} \geq 0$.
3. In each row of the regret matrix, take the maximum.
4. Choose the action with the **smallest** of these maxima:

$$\min_{a_i} \left\{ \max_{\theta_j} r_{ij} \right\}$$

r_{ij} : how much is lost by having chosen a_i , **finding out afterwards** that θ_j occurred.

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	θ_1	θ_2	θ_3
a_1	40	45	50
a_2	20	60	80
a_3	-40	50	120
max.	40	60	120

gains matrix

→

	θ_1	θ_2	θ_3	max.
a_1	0	15	70	70
a_2	20	0	40	40
a_3	80	10	0	80

regret matrix

$$\min\{70, 40, 80\} = 40 \Rightarrow a_2 \text{ (medium)}$$

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	θ_1	θ_2	θ_3		θ_1	θ_2	θ_3	max.	
a_1	40	45	50	$\longrightarrow$	a_1	0	15	70	70
a_2	20	60	80		a_2	20	0	40	40
a_3	-40	50	120		a_3	80	10	0	80
max.	40	60	120						
gains matrix					regret matrix				

$$\min\{70, 40, 80\} = 40 \Rightarrow a_2 \text{ (medium)}$$

It is a **pessimistic** criterion (“min–max”), but applied to the regrets and not to the gains: that is why it does not choose the same as Maximin.

The Savage criterion can be inconsistent

Two actions, two states:

	A	B		A	B	max.
Action 1	60	-40	→	10	0	10
Action 2	70	-60		0	20	20

$$\min\{10, 20\} = 10 \Rightarrow \text{Action 1}$$

The Savage criterion can be inconsistent

Two actions, two states:

	A	B		A	B	max.
Action 1	60	-40	→	10	0	10
Action 2	70	-60		0	20	20

$$\min\{10, 20\} = 10 \Rightarrow \text{Action 1}$$

A third action is added, **keeping the others**:

	A	B		A	B	max.
Action 1	60	-40	→	30	0	30
Action 2	70	-60		20	20	20
Action 3	90	-65		0	25	25

$$\min\{30, 20, 25\} = 20 \Rightarrow \text{Action 2}$$

The Savage criterion can be inconsistent

Two actions, two states:

	A	B		A	B	max.
Action 1	60	-40	→	10	0	10
Action 2	70	-60		0	20	20
$\min\{10, 20\} = 10 \Rightarrow \text{Action 1}$						

A third action is added, **keeping the others**:

	A	B		A	B	max.
Action 1	60	-40	→	30	0	30
Action 2	70	-60		20	20	20
Action 3	90	-65		0	25	25
$\min\{30, 20, 25\} = 20 \Rightarrow \text{Action 2}$						

What happened?

Action 3 **is not chosen**, but its presence reversed the preference between Actions 1 and 2: by changing the maximum of column A, it changed all the regrets in that column. Criteria I, II, III and V do not have this problem: the value of each row depends only on that row.

V. Hurwicz criterion (moderate pessimism)

V. Hurwicz, with coefficient of optimism $\alpha \in [0, 1]$

For each action:

- expected optimism: $\alpha \cdot \max_j c_{ij}$
- expected pessimism: $(1 - \alpha) \cdot \min_j c_{ij}$

$$\max_{a_i} \left\{ \alpha \max_{\theta_j} c_{ij} + (1 - \alpha) \min_{\theta_j} c_{ij} \right\}$$

$\alpha = 1$: Maximax. $\alpha = 0$: Maximin.

The decision maker chooses α — it is their attitude towards risk.

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With $\alpha = 0.5$:

	$0.5 \times \max$	$0.5 \times \min$	sum
a_1	$0.5 \times 50 = 25$	$0.5 \times 40 = 20$	45
a_2	$0.5 \times 80 = 40$	$0.5 \times 20 = 10$	50
a_3	$0.5 \times 120 = 60$	$0.5 \times (-40) = -20$	40

$$\max\{45, 50, 40\} = 50 \Rightarrow a_2 \text{ (medium)}$$

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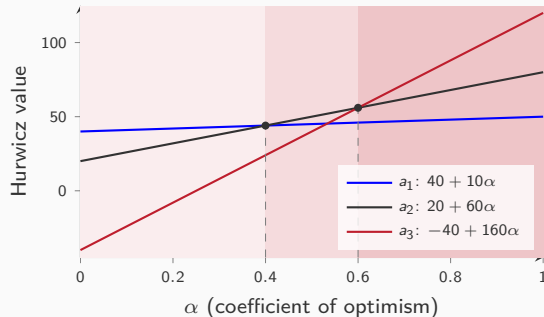
With $\alpha = 0.5$:

	$0.5 \times \max$	$0.5 \times \min$	sum
a_1	$0.5 \times 50 = 25$	$0.5 \times 40 = 20$	45
a_2	$0.5 \times 80 = 40$	$0.5 \times 20 = 10$	50
a_3	$0.5 \times 120 = 60$	$0.5 \times (-40) = -20$	40

$$\max\{45, 50, 40\} = 50 \Rightarrow a_2 \text{ (medium)}$$

Question: and with $\alpha = 0.7$? (*More optimistic: 47, 62, 72 — it switches to a_3 .*)

Hurwicz: the decision depends on α



Each action gives a **straight line** in α ; we choose the one **on top**.

α	decision
$[0, 0.4[$	a_1 (small)
$]0.4, 0.6[$	a_2 (medium)
$]0.6, 1]$	a_3 (large)

The crossing points:

$$40 + 10\alpha = 20 + 60\alpha \iff \alpha = 0.4;$$

$$20 + 60\alpha = -40 + 160\alpha \iff \alpha = 0.6.$$

A single value of α hides this: it is worth knowing from what level of optimism the decision changes.

Five criteria, three decisions

Criterion	attitude	decision
I. Maximax	optimistic	a_3 large
II. Maximin	pessimistic	a_1 small
III. Laplace	"neutral"	a_2 medium
IV. Savage	regret	a_2 medium
V. Hurwicz ($\alpha = 0.5$)	moderate	a_2 medium

There is no "right" criterion

Depending on the situation, one criterion may be more suitable than another, although each one has its weaknesses. Using several criteria **simultaneously** is also useful for whoever has to decide.

Here, the medium line is chosen by three different criteria: it is a decision that is **robust** to the choice of criterion.

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What if we have some information about the probability of each state? → 1.3 Probabilistic criteria.

What if the matrix contains costs?

If c_{ij} is a **cost** (or loss), the smaller the better: **just reverse the directions**.

Criterion	gains (maximise)	costs (minimise)
I. optimistic	$\max_i \max_j c_{ij}$ (Maximax)	$\min_i \min_j c_{ij}$ (Minimin)
II. pessimistic	$\max_i \min_j c_{ij}$ (Maximin)	$\min_i \max_j c_{ij}$ (Minimax)
III. Laplace	$\max_i \frac{1}{m} \sum_j c_{ij}$	$\min_i \frac{1}{m} \sum_j c_{ij}$
IV. Savage	$r_{ij} = \max_k c_{kj} - c_{ij}$	$r_{ij} = c_{ij} - \min_k c_{kj}$
	in both cases: $\min_i \max_j r_{ij}$	
V. Hurwicz	$\max_i \{\alpha \max_j c_{ij} + (1 - \alpha) \min_j c_{ij}\}$	$\min_i \{\alpha \min_j c_{ij} + (1 - \alpha) \max_j c_{ij}\}$

The regret is always ≥ 0 and measures the distance to the best action in that column. In Hurwicz, α is still the weight of the **best** result — which, with costs, is the minimum.

Example 3 — investment

Actions	market conditions		
	rise	stable	fall
buy shares	1000	0	-1500
buy bonds	350	200	300
buy treasury bills	220	100	0

Which action should be chosen according to the Maximax, Maximin, Laplace, Savage and Hurwicz (coefficient of optimism 0.8) criteria?

Exercise — estimated gains (m.u.)

Actions	θ_1	θ_2	θ_3
a_1	-50	0	80
a_2	-10	30	35
a_3	60	45	-30
a_4	90	40	45

Which of the actions should we choose according to:
a) Maximax; b) Maximin; c) Laplace; d) Savage; e) Hurwicz with coefficient of optimism 0.7?

Before starting: is any action dominated?

KEY TAKEAWAYS — 1.2

1. Without probabilities, each criterion **summarises each row** of the matrix in one number and chooses the best row; dominated actions are eliminated first.
2. **Maximax** (optimistic) and **Maximin** (pessimistic) look only at the best and the worst result of each action.
3. **Laplace**: equally likely states — uses the whole row, but depends on how the states are defined.
4. **Savage**: minimises the maximum **regret**; adding an action that is not chosen can change the decision.
5. **Hurwicz**: $\alpha \max + (1 - \alpha) \min$, between Maximin ($\alpha = 0$) and Maximax ($\alpha = 1$); the decision changes with α .
6. With **costs**, the directions are reversed.

Different criteria, different decisions: **the criterion is the decision maker's choice**.

What if we know something about the probability of each state? → 1.3 Probabilistic criteria.

Exercise sheet 1 — Exercise 1 (homework)

The students' union game

The students' union organises the following game: 30 white balls, 20 black balls and 10 red balls are placed in a black bag; the player chooses game A or game B and draws a ball at random. Gains (€):

Decision	White	Black	Red
Game A	-20	-5	100
Game B	-5	5	50

Which game do you decide to play, according to the criteria a) Maximax; b) Maximin; c) Laplace; d) Savage; e) Hurwicz with coefficient of optimism 0.7?

f) (After 1.3.) Use the contents of the bag to assign a probability to each colour. Which game do you choose by expected value? Compare with c).

Submission: by email to the lecturer, by the deadline given in class. All exercises: [Exercise sheet 1](#).

Exercise sheet 1 — Exercise 2 (homework)

Maintenance plan for a fleet (cost matrix)

A transport company is choosing the maintenance plan for its fleet for next year. The total annual **cost** (maintenance plus repairs, in thousands of euros) depends on the number of breakdowns that occur:

Plan	few breakdowns	moderate breakdowns	many breakdowns
corrective (repair only)	10	40	95
quarterly preventive	25	35	60
monthly preventive	45	45	50
external contract	35	40	65

a) Is any plan dominated? Which plan should be chosen according to the criteria b) optimistic; c) pessimistic; d) Laplace; e) Savage; f) Hurwicz with coefficient of optimism 0.6?

Submission: by email to the lecturer, by the deadline given in class.

Exercise sheet 1 — Exercise 3 (homework)

How much to order?

A shop has to order q units of a seasonal product **before** knowing the season's demand d . Each unit costs 6 € and sells for 10 €; units that are not sold are cleared at the end of the season at 3 € each. The shop assumes that demand will be 100, 200, 300 or 400 units, and only orders in batches of 100.

- a) Build the decision matrix: actions $q \in \{100, 200, 300, 400\}$, states $d \in \{100, 200, 300, 400\}$, consequences: profit for the season (€).
- b) How much should be ordered according to the Maximax, Maximin, Laplace, Savage and Hurwicz criteria with coefficient of optimism 0.7?
- c) What attitude towards risk corresponds to each of the quantities chosen?

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References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Decision Analysis.
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017 — decision under uncertainty (Laplace, minimax, Savage, Hurwicz).
- Milnor, J., *Games against nature*, in R. M. Thrall, C. H. Coombs, R. L. Davis (eds.), *Decision Processes*, Wiley, 1954, pp. 49–59 — the properties of each criterion (including the inconsistency of the Savage criterion).